C.W

05.10.2026

(*) Last math (2nd Part)

$$A=\begin{bmatrix}1 & 4\\ 9 & 1\end{bmatrix}, \lambda=-5,7, V_1=\begin{pmatrix}-2\\3\end{pmatrix}, V_2=\begin{pmatrix}2\\3\end{pmatrix}$$

find P^{-1} such that $P^{-1}AP$ = diagonal matrix.

$\Rightarrow$ Let, V_1 V_2

$$P=\begin{bmatrix}-2 & 2\\ 3 & 3\end{bmatrix}$$

$$|P| = \begin{vmatrix}-2 & 2\\ 3 & 3\end{vmatrix} = -6-6 = -12$$

cofactors:

$a_{11} = 3$

$a_{12} = -3$

$a_{21} = -2$

$a_{22} = -2$

$\therefore$ cofactor matrix $= \begin{bmatrix} 3 & -3 \\ -2 & -2 \end{bmatrix}$

$\therefore$ Adj of $(P) = \begin{bmatrix} 3 & -3 \\ -2 & -2 \end{bmatrix}^T$

$= \begin{bmatrix} 3 & -2 \\ -3 & -2 \end{bmatrix}$

$\therefore P^{-1} = \frac{1}{-12} \begin{bmatrix} 3 & -2 \\ -3 & -2 \end{bmatrix} = \frac{1}{12} \begin{bmatrix} -3 & 2 \\ 3 & 2 \end{bmatrix}$

$\therefore P^{-1}AP$

$= \frac{1}{12} \begin{bmatrix} -3 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 9 & 1 \end{bmatrix} \begin{bmatrix} -2 & 2 \\ 3 & 3 \end{bmatrix}$

$= \frac{1}{12} \begin{bmatrix} -3+18 & -12+2 \\ 3+18 & 12+2 \end{bmatrix} \begin{bmatrix} -2 & 2 \\ 3 & 3 \end{bmatrix}$

$= \frac{1}{12} \begin{bmatrix} 15 & -10 \\ 21 & 14 \end{bmatrix} \begin{bmatrix} -2 & 2 \\ 3 & 3 \end{bmatrix}$

$= \frac{1}{12} \begin{bmatrix} -30-30 & 30-30 \\ -42+42 & 42+42 \end{bmatrix}$

$= \frac{1}{12} \begin{bmatrix} -60 & 0 \\ 0 & 84 \end{bmatrix}$

$= \begin{bmatrix} -5 & 0 \\ 0 & 7 \end{bmatrix}$

CamScanner